

Project Documentation: Smart Emergency Wearable for Women & Elderly

Project Title:

SHEILD – Smart Health and Emergency Instant-Location Device

Overview

SHEILD is a compact, wearable emergency device developed for **women's safety** and **elderly care**, combining health monitoring, SOS signaling, and location broadcasting — with both **online (GSM/WiFi)** and **offline (LoRa/Bluetooth Mesh)** communication capabilities.

It is designed to be **affordable**, **discreet**, and **easy to use**, even in panic or unconscious states.

Objectives

- Enable **emergency alerts** via a wearable device using **panic buttons**.
 - Provide **location tracking**, **offline messaging**, and **audio deterrents**.
 - Store **essential medical information** for elderly users.
 - Facilitate **community-based emergency response** even without internet.
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System Components

a. Hardware Components

Component	Function
ESP32 / Arduino Nano	Main microcontroller
GSM Module (SIM800L) or WiFi	Sends SMS/location alerts
Pulse Sensor (MAX30100)	Detects heart rate anomalies
Buzzer & LED/Strobe	Deterrence / alerting nearby people
Lithium Battery (3.7V)	Portable power supply
LoRa Module (SX1278)	Offline mesh alert system
Bluetooth	Local phone pairing & control
EEPROM/Flash Memory	Store user medical profile
Enclosure (Pendant/Bracelet/Keychain)	Discreet wearable form

b. Software Functions

- **Trigger Modes:**
 - Long-press panic button
 - Fall detection via accelerometer
 - Heart rate irregularity detection
- **Alert Features:**
 - Send SMS with GPS coordinates
 - Activate buzzer/strobe
 - Optional: Transmit ambient audio (evidence)
- **Offline Communication:**
 - LoRa Mesh: Broadcast alert to nearby responders
 - Bridgefy/FireChat App: Phone-based mesh messaging
- **Elderly Support:**
 - Stores emergency data (e.g., blood group, conditions)
 - Sends SOS to caregivers, paramedics
- **Mobile App:**
 - Configuration of emergency contacts
 - View device status
 - Update user's medical information

Use Case Scenarios

1. Women's Safety

- A woman walking home feels threatened → Presses panic button → Loud alarm triggers → SMS sent to contacts with live GPS.

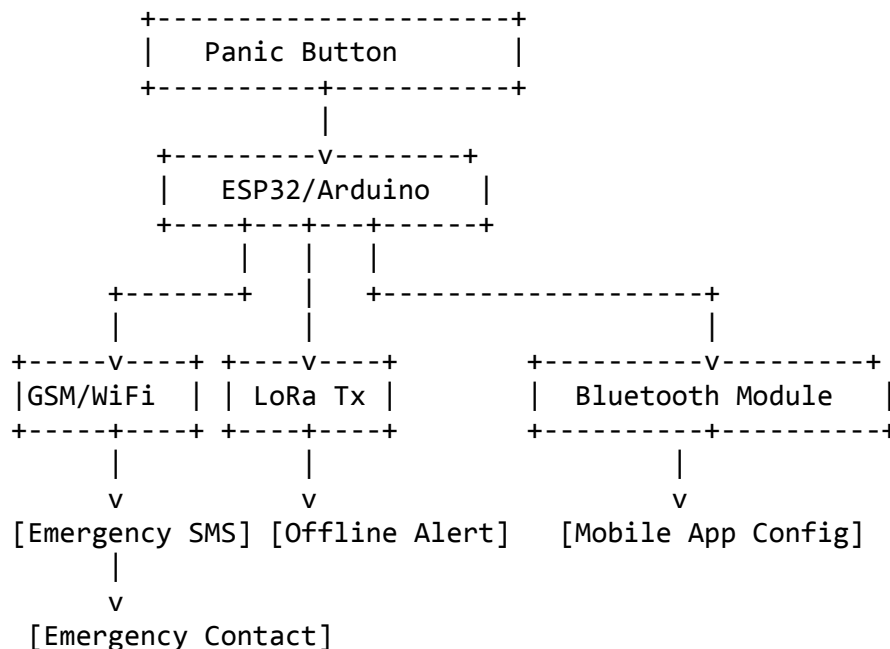
2. Elderly Care

- An elderly person collapses due to a fall → Fall is auto-detected → Sends SOS + medical profile to family & ambulance.

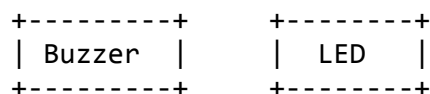
3. No Internet Coverage

- In a disaster zone with no network → Alert transmitted via LoRa to nearby users → Community responds.
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Block Diagram



Others:



Estimated Cost (Per Unit Prototype)

Component	Approx Cost (INR)
ESP32	₹300
GSM Module	₹400
MAX30100	₹250
MPU6050	₹200
LoRa Module	₹350
Lithium Battery	₹150
Buzzer + LED	₹50
Misc (PCB, Enclosure, Wiring)	₹300
Total (Prototype)	XXXXXX

Mobile App Features

- Contact management (trusted numbers)
 - Emergency data input (medical history)
 - Alert log
 - Bluetooth control interface
 - Toggle sensor features (fall, heart rate, etc.)
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Workflow Summary

- 1. Initialization**
 - Device powered on, connects to GSM/WiFi or LoRa
 - Medical data loaded from internal memory
 - 2. Monitoring Mode**
 - Constantly checks sensor values (heart rate, acceleration)
 - Waits for panic button press or automatic trigger
 - 3. Trigger Event**
 - Alert signal sent via GSM/LoRa
 - Sound buzzer + flash LED
 - Mobile app displays alert status
 - 4. Post-Event**
 - Logs data
 - Resets after confirmation
 - Optional: Sends audio to app/cloud
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Future Enhancements

- GPS tracker with geofencing
 - AI-based anomaly detection in health data
 - Voice-controlled panic command
 - Solar-powered version
 - Auto sync with medical databases or hospitals
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Conclusion

This wearable emergency system aims to **empower women, protect the elderly, and strengthen community-based rapid response** — even in low-connectivity scenarios. With simple hardware and intelligent software integration, it provides a **lifesaving solution** that's accessible, portable, and powerful.